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Department of Electronics and Communication Engineering



Project Report on

Smart Glove for paralysed people

Submitted in partial fulfillment of the requirement for the award of the degree of

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in

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ABSTRACT

Paralysed patients often face difficulty in performing basic hand movements and communicating immediate discomfort such as high temperature, dryness, or moisture imbalance in their limbs. To address these challenges, this project presents a smart Assistive Glove System developed using an ESP32 microcontroller. The glove integrates a temperature sensor (LM35), a moisture sensor, a servo-based finger movement mechanism, an LCD display, and a buzzer alert unit. The system continuously monitors the patient's hand temperature and moisture level, displaying real-time data on the LCD. When the temperature exceeds a safe limit or the moisture level drops below a threshold, the system automatically activates alerts through a buzzer and stops all servo movement for safety.

A smooth 0°–90° servo motion is implemented using ESP32-based PWM control to assist limited finger movement, promoting mild exercise and reducing stiffness. A dedicated ON/OFF control button ensures that caregivers can enable or disable the system as needed. The glove operates on a compact battery system, making it lightweight, portable, and suitable for continuous use.

This smart glove provides an effective assistive tool to improve patient comfort, enhance safety, and support basic rehabilitation for individuals with paralysis. It is designed to be cost-effective, easy to operate, and adaptable for further IoT-based monitoring and medical applications.

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FLOW OF THE PROJECT

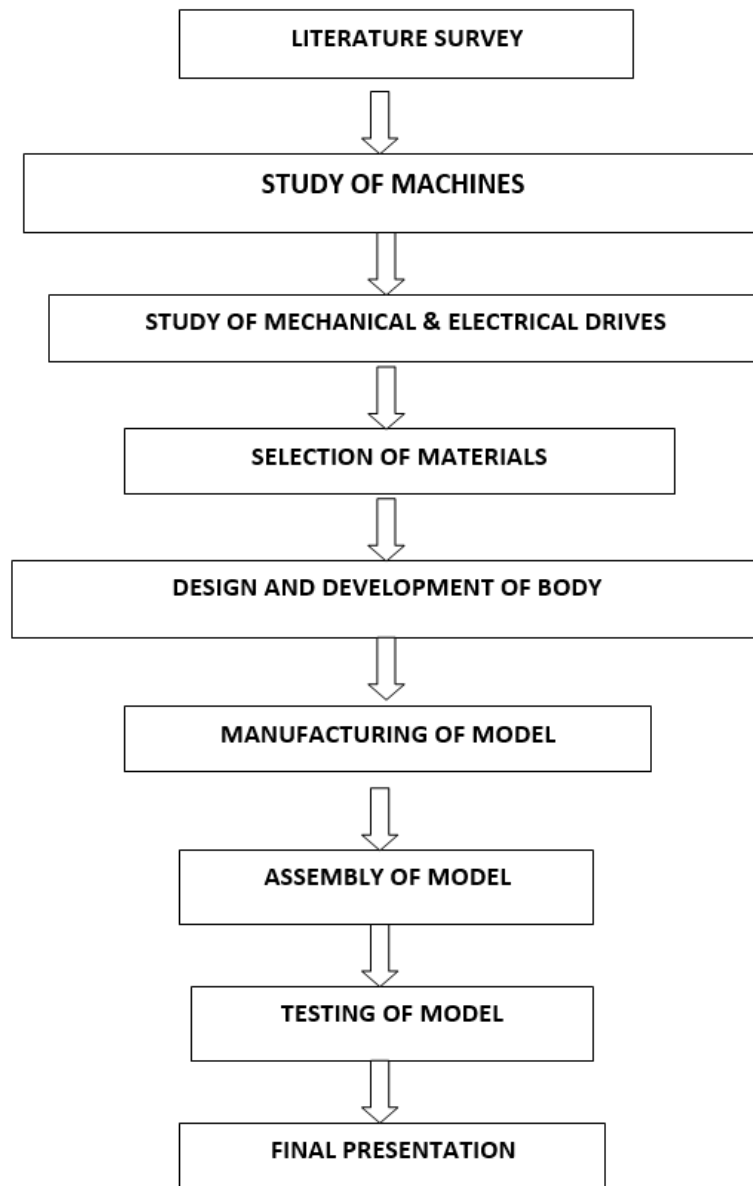


Table 1: Flow of the project

CHAPTER 1—INTRODUCTION

Paralysis is a medical condition that results in the loss of muscle function in one or more parts of the body, often caused by stroke, spinal cord injury, nerve damage, or neurological disorders. Individuals suffering from paralysis face severe challenges in performing basic daily activities and require continuous support from caregivers. One of the major issues observed in paralysed limbs is the inability to sense and respond to changes in temperature, moisture, or dryness, which may lead to discomfort, skin infections, muscle stiffness, and reduced blood circulation. Additionally, lack of voluntary movement in the fingers or hand can further weaken the muscles over time.

To address these challenges, assistive wearable technologies offer an effective solution. This project introduces a Smart Assistive Glove System designed specifically for paralysed patients, aimed at providing safety monitoring, mild rehabilitation, and caregiver assistance. Using an ESP32 microcontroller, the glove integrates essential sensors such as LM35 temperature sensor and moisture sensor to continuously track the patient's hand condition. It also includes a servo motor mechanism to support controlled finger movement, which helps prevent stiffness and encourages light physiotherapy exercises.

The system displays real-time readings on an LCD screen and activates a buzzer when unsafe conditions such as high temperature or low moisture are detected. The glove is battery-powered, lightweight, and portable, making it suitable for home, rehabilitation centers, and hospital use. A dedicated ON/OFF button ensures easy operation for both patients and caregivers.

Overall, the Smart Assistive Glove acts as a supportive medical aid that enhances patient comfort, improves safety monitoring, and provides a practical rehabilitation tool to improve the quality of life for individuals with paralysis.

CHAPTER 2 — LITERATURE SURVEY

Paper 1

Title: *Wearable Assistive Exoskeleton Glove for Hand Rehabilitation* Authors: Y. Zheng, L. Wang, M. Koivuniemi Year: 2022

Abstract:

This paper presents a soft robotic glove designed to assist stroke and paralysis patients in rehabilitating their fingers. The glove uses servo motors and tendon-based mechanisms to generate finger motion. Flexible sensors provide real-time feedback on user movement. The system improves hand mobility and offers cost-effective rehabilitation.

Limitations:

- Limited grip strength for fully paralyzed patients.
- High power consumption reduces battery runtime.
- Not optimized for continuous daily use.

Paper 2

Title: *Smart Glove for Hand Movement Assistance Using Embedded Sensors*

Authors: S. Hussain, T. Ahmad

Year: 2021

Abstract:

The study introduces a smart glove utilizing flex sensors, temperature sensors, and microcontrollers to help patients regain hand coordination. The glove detects finger bending and provides controlled motion using actuators. The system includes warning alerts for abnormal temperature and dryness.

Limitations:

- Movement accuracy depends heavily on sensor calibration.
- Flex sensors degrade with prolonged use.
- Limited IoT capability.

Paper 3

Title: *Servo-Based Soft Exoskeleton for Post-Stroke Hand Therapy*

Authors: Q. Sheng, M. Li

Year: 2020

Abstract:

This paper provides a design methodology for servo-driven rehabilitation exoskeleton gloves.

The device uses nylon-tendon transmission to mimic natural finger flexion. Experimental results show improved rehabilitation outcomes with repetitive finger motion cycles.

Limitations:

- Bulkier than expected due to servo mounting.
- Noise and vibration from servos reduce comfort.
- Lacks advanced user safety features like temperature monitoring.

Paper 4

Title: *IoT-Based Wearable Health Monitoring System*

Authors: Pallavi C., Arun Kumar

Year: 2020

Abstract:

This research proposes an IoT-enabled wearable system for continuous monitoring of vital parameters such as temperature, moisture level, and pulse rate. The system uses ESP32 and cloud platforms for real-time data logging and emergency alerts.

Limitations:

- Not specifically designed for hand rehabilitation.
- Moisture sensors show inconsistent readings with sweat.
- IoT connectivity sometimes disconnects due to Wi-Fi instability

Paper 5

Title: *Flexible Sensors and Wearable Electronics for Healthcare Applications*

Authors: J. Park, H. Lee

Year: 2020

Abstract:

The paper reviews advances in flexible and wearable sensors used in modern healthcare devices. It emphasizes temperature monitoring, moisture detection, and patient comfort. The study also explores integration challenges involving power supply, safety, and long-term wearability.

Limitations:

- Mainly theoretical; lacks a complete assistive glove system.
- Does not include servo-actuated rehabilitation solutions.
- Limited details on patient trials.

CHAPTER 3 — PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Paralysed patients experience limited or no voluntary movement in their hands and fingers, making them unable to sense or respond to changes in temperature, moisture, or dryness. This lack of sensitivity and movement often leads to complications such as skin infections, tissue damage, muscle stiffness, and poor blood circulation. Caregivers must constantly monitor the patient's limb condition, which is time-consuming and not always feasible.

There is currently no low-cost, wearable system that can simultaneously monitor hand temperature and moisture, provide automatic alerts, and assist in basic finger movement rehabilitation for paralysed individuals. Therefore, there is a need for a compact, intelligent, and user-friendly device that can support continuous monitoring, ensure patient safety, and improve hand mobility through controlled movement.

Objectives

1. To design a wearable assistive glove that provides basic hand movement support and real-time monitoring for paralysed patients.
2. To measure hand temperature using the LM35 sensor and detect abnormal levels that may lead to discomfort or infection.
3. To monitor hand moisture levels to prevent dryness-related skin issues or high-moisture infections.
4. To generate automatic safety alerts using a buzzer whenever unsafe temperature or moisture conditions are detected.
5. To assist finger movement through a controlled servo motor mechanism for mild physiotherapy and prevention of muscle stiffness.
6. To develop a compact, portable, battery-operated system with LCD display and ON/OFF control for easy use in homes, hospitals, and rehabilitation centers.

CHAPTER 4 — PROPOSED METHODOLOGY

The Smart Assistive Glove is developed using a systematic and structured engineering workflow that ensures accuracy, reliability, and user comfort. The proposed methodology includes the following major stages:

1. Problem & Requirement Analysis

Understanding challenges faced by paralysed patients such as loss of sensation, inability to detect temperature/moisture, and limited finger movement.

Identifying the need for real-time monitoring, safety alerts, and motor-assisted finger movement.

Defining system requirements such as lightweight design, portability, battery operation, and user-friendly interface.

2. Conceptual Design & Planning

Designing the architecture including sensors, actuator (servo), ESP32 controller, LCD display, and buzzer.

Choosing components considering compactness, low power consumption, and ease of integration.

Planning placement of sensors and servo on the glove for maximum comfort and accuracy.

3. Hardware Integration

Connecting LM35, moisture sensor, servo motor, LCD, buzzer, and ON/OFF switch to ESP32.

Mounting components securely onto the glove while ensuring flexibility and comfort.

Managing power distribution using a 3.7V battery pack and voltage regulation.

4. Software Implementation

Programming ESP32 using Arduino IDE.

Writing code for sensor reading, LCD display, servo control, and alert triggering.

Adding safety features such as system halt and buzzer activation during unsafe conditions.

5. Calibration & Threshold Setting

Calibrating temperature and moisture sensors under real conditions.

Setting safe operating thresholds and fine-tuning servo movement angles.

6. Testing & Validation

Testing each module independently and then as a complete system.

Checking performance in different temperature and moisture conditions.

Verifying responsiveness, comfort, and battery performance.

7. Final Optimization & Documentation

Improving ergonomics, code efficiency, and power stability.

Preparing final documentation with results, diagrams, and analysis.

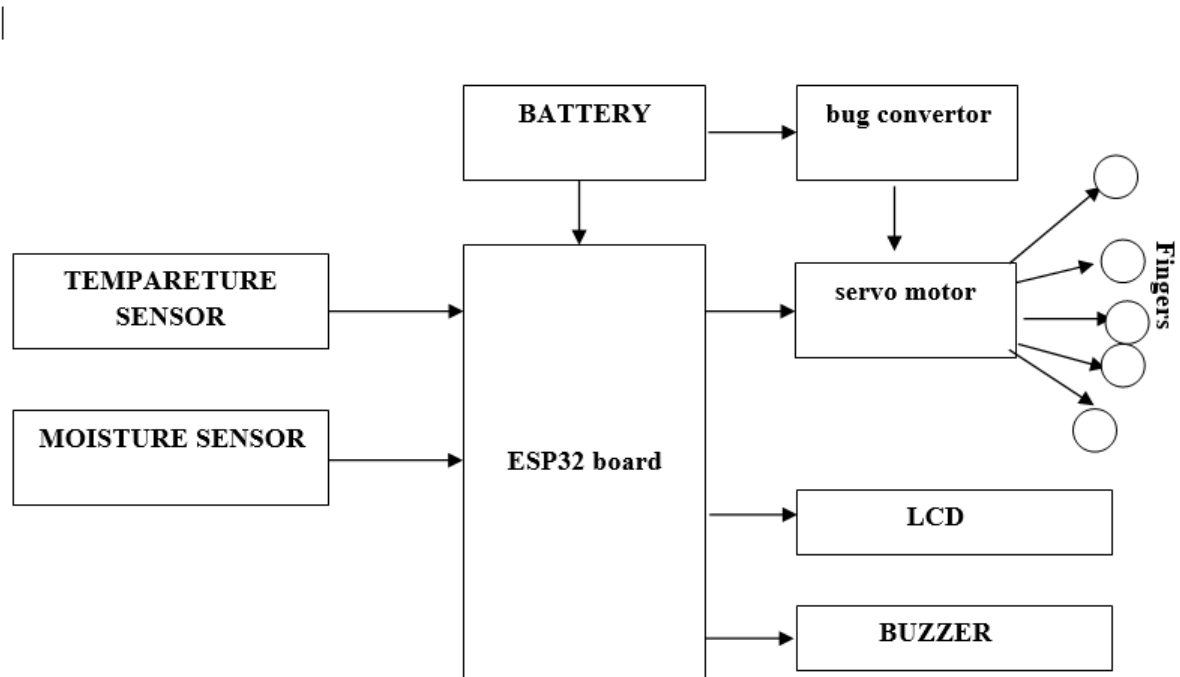


Figure 1: Block Diagram of Smart Assistive Glove System

CHAPTER 5 — HARDWARE & SOFTWARE COMPONENTS

1.ESP32 board – A powerful microcontroller with Wi-Fi and Bluetooth used to control and connect IoT devices.



Figure 2: ESP32 Microcontroller Pin Diagram

2. Servo motor – A precision motor that rotates to a specific angle based on the control signal.



Figure 3: Servo Motor Mechanism for Finger Movement

3. Buck converter (12V to 5V for servo) – A DC-DC module that steps down 12V input to a stable 5V output for powering the servo motor safely.

4. Jumper wires – Wires used to make quick electrical connections between components on a breadboard or modules.

5. 3.7V*3 Lithium battery – Rechargeable lithium battery pack used as a compact and reliable power source.

- 6. Foam sheet** – A lightweight material used for mounting, insulation, or structural support in the project.
- 7. DC pin for wire fitting** – A connector pin used to attach and supply DC power to the circuit.
- 8. ESP32 expander** – A module that increases the number of input/output pins available for the ESP32.
- 9. Nylon wire** – A strong, flexible thread used for mechanical movement or pulling mechanisms.
- 10. Gloves** – Protective gloves used for safe handling of components and assembly.
- 11. Battery holder** – A casing that securely holds the battery and provides electrical terminals for connection.
- 12. LCD display with I2C** – A screen that shows text or data and uses the I2C interface to reduce required wiring.
- 13. LM35 temperature sensor** – A sensor that measures temperature and outputs an analog voltage proportional to the heat.
- 14. Moisture sensor** – A sensor that detects the moisture level in soil or the environment.
- 15. Button for ON/OFF** – A simple switch used to manually turn the device on or off.
- 16. Buzzer** – An audio component that produces a sound or alert signal.

Component	Type	Key Role	Interface	Code Pin
ESP32	Microcontroller	Central Controller/Processor	Digital/Analog/I ² C	N/A
LM35	Sensor	Temperature Monitoring	Analog	<i>GPIO32</i>
Moisture Sensor	Sensor	Wetness Detection	Analog	<i>GPIO34</i>
Servo Motor	Actuator	Assisted Finger Movement	PWM	<i>GPIO17</i>
16×2 LCD	Output Module	Real-time Data Display	I ² C (SDA/SCL)	0x27
Buzzer	Output Module	Audible Warning Alert	Digital Output	<i>GPIO16</i>
Battery/Regulator	Power Supply	System Power Source	Power Lines	VIN/GND

Table 2: Specifications of Sensors and Modules.

CHAPTER 6— SYSTEM DESIGN & WORKING PRINCIPLE

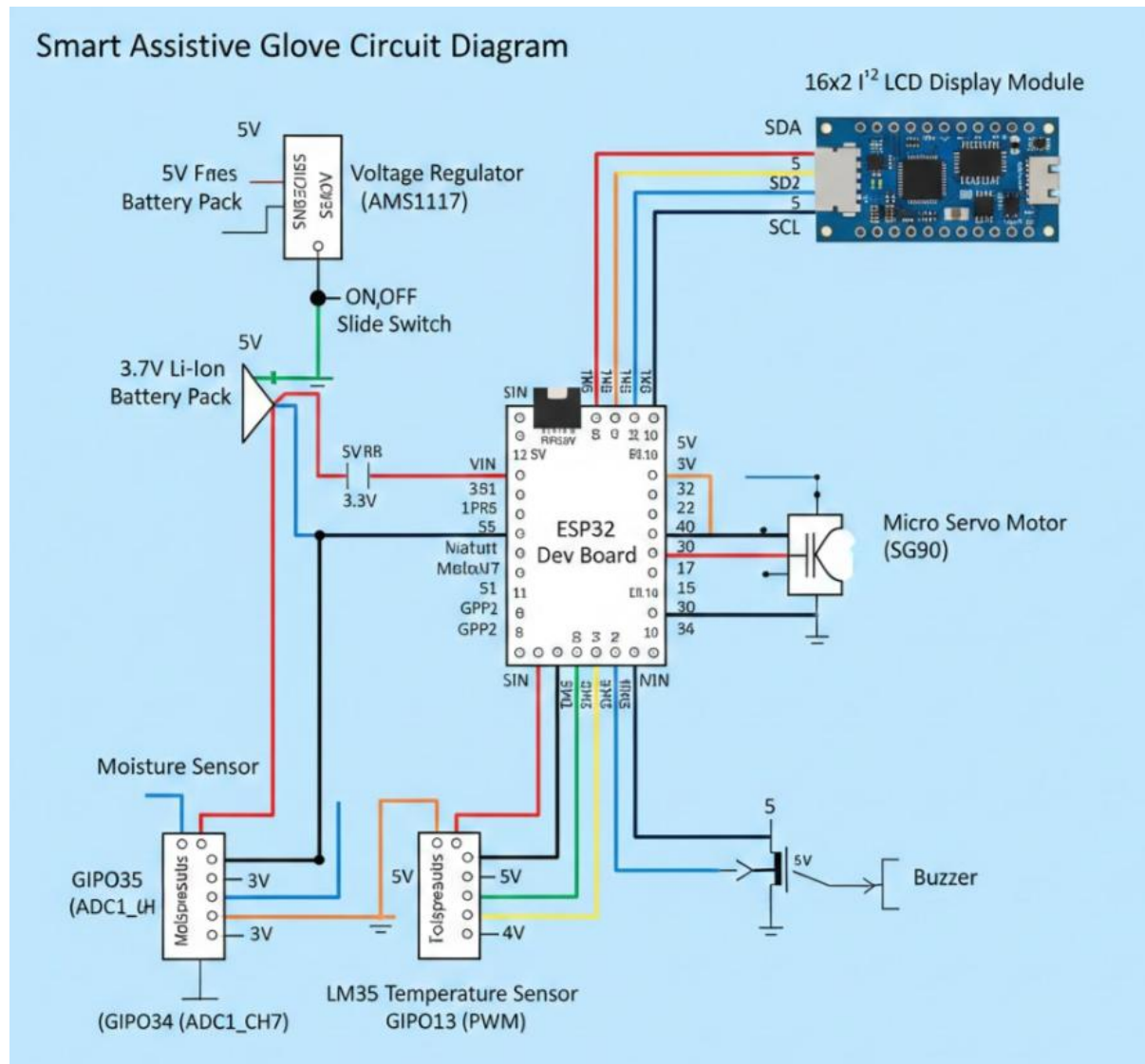


Figure 4: Circuit Diagram of the Proposed System

1. Power ON

When the ON/OFF button is pressed, the ESP32 gets power from the battery.

The buck converter ensures that the servo receives a safe 5V supply.

2. Sensor Data Collection

The LM35 sensor continuously measures temperature and sends analog voltage to the ESP32.

The Moisture sensor checks the soil moisture level and sends its reading to the controller.

3. Processing in ESP32

The ESP32 reads the sensor values, converts them to digital format, and processes them.

Based on programmed threshold values, it decides what action to take.

4. Decision and Control

If temperature is high or moisture is low, the ESP32 triggers specific actions:

The servo motor performs a scheduled motion (e.g., opening/closing a flap, moving an arm).

The buzzer rings if any reading is out of safe limits. If readings are normal, the system stays idle and conserves power.

5. Display Output

The LCD (I2C) shows real-time values such as:

Temperature (°C)

Moisture level (%)

System status (Normal / Alert / Servo Active)

Parameter	Sensor	Variable Name in Code	Threshold Value	Notes from Code/Design
Temperature	LM35	TEMP_THRESHOLD	35.0°C	Temperature above this triggers an alert. LM35 converts raw ADC to Celsius.
Moisture	Moisture Sensor	MOISTURE_HIGH_THRESHOLD	2500 (Raw ADC)	Raw ADC value above this triggers an alert. ESP32 uses 12-bit ADC (Max 4095).
Servo Position 1	Servo Motor	SERVO_POS1	0°	Represents the starting/resting position of the finger.
Servo Position 2	Servo Motor	SERVO_POS2	90°	Represents the maximum assisted flexed position.
Rehabilitation Cycles	System Control	CYCLES	10	The number of times the 0° ↔ 90° motion will be performed.

6. Mechanical Action

The servo movement is transmitted using nylon wire, especially if the motion needs extending or pulling.

The foam sheet provides the base for fixing sensors, display, and components.

7. Continuous Monitoring

The system keeps running until the user turns OFF the device using the push button.

CHAPTER 7 - COST EXPENDITURE

SL No	Particulars	Cost in Rs
1	ESP32 board	250
2	battery	150
3	Servo motor	159
4	Jumper wire	50
5	ESP32 expander	100
6	Temperature sensor	200
7	Moisture sensor	50
8	buzzer	20
9	plywood	300
10	Bug convertor 12v to 5v	120
11	Form sheet	50
12	LCD display	150
13	I2c	100
14	Nylon wire	100
15	Button	20
16	Battery holder	30
17	Glove	50
18	DC pin	10

Table 5: Cost expenditure

CHAPTER 8 -PROTOTYPE

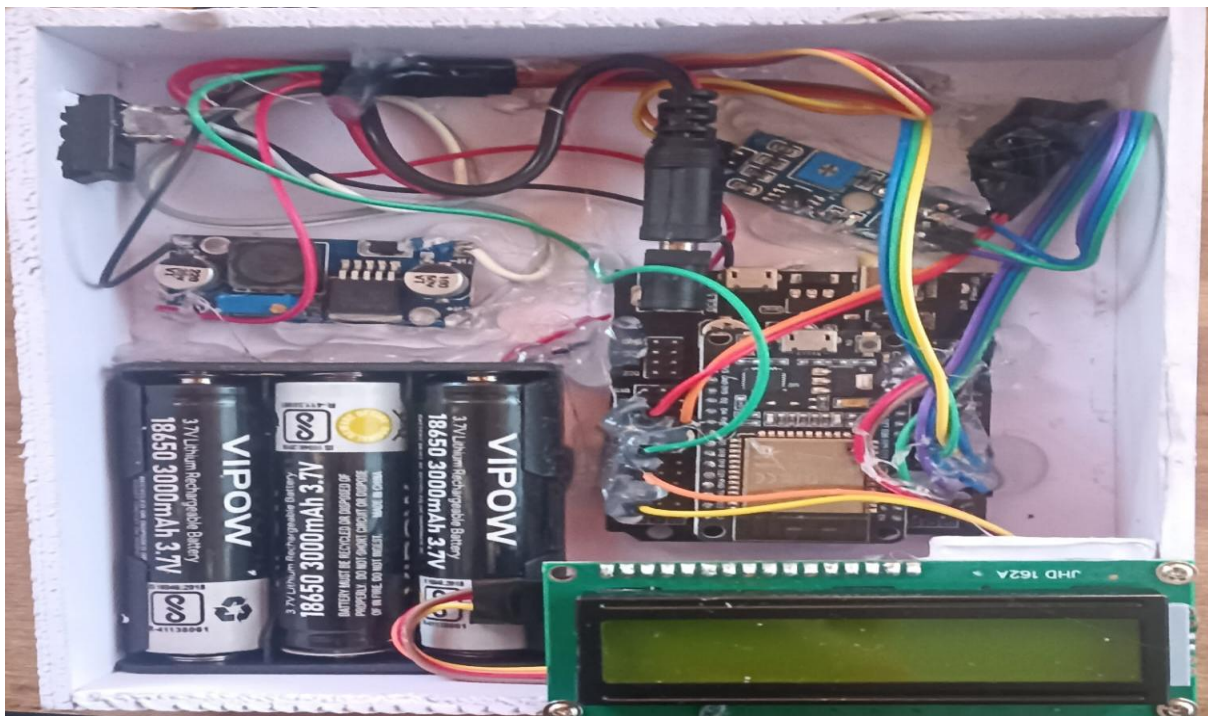
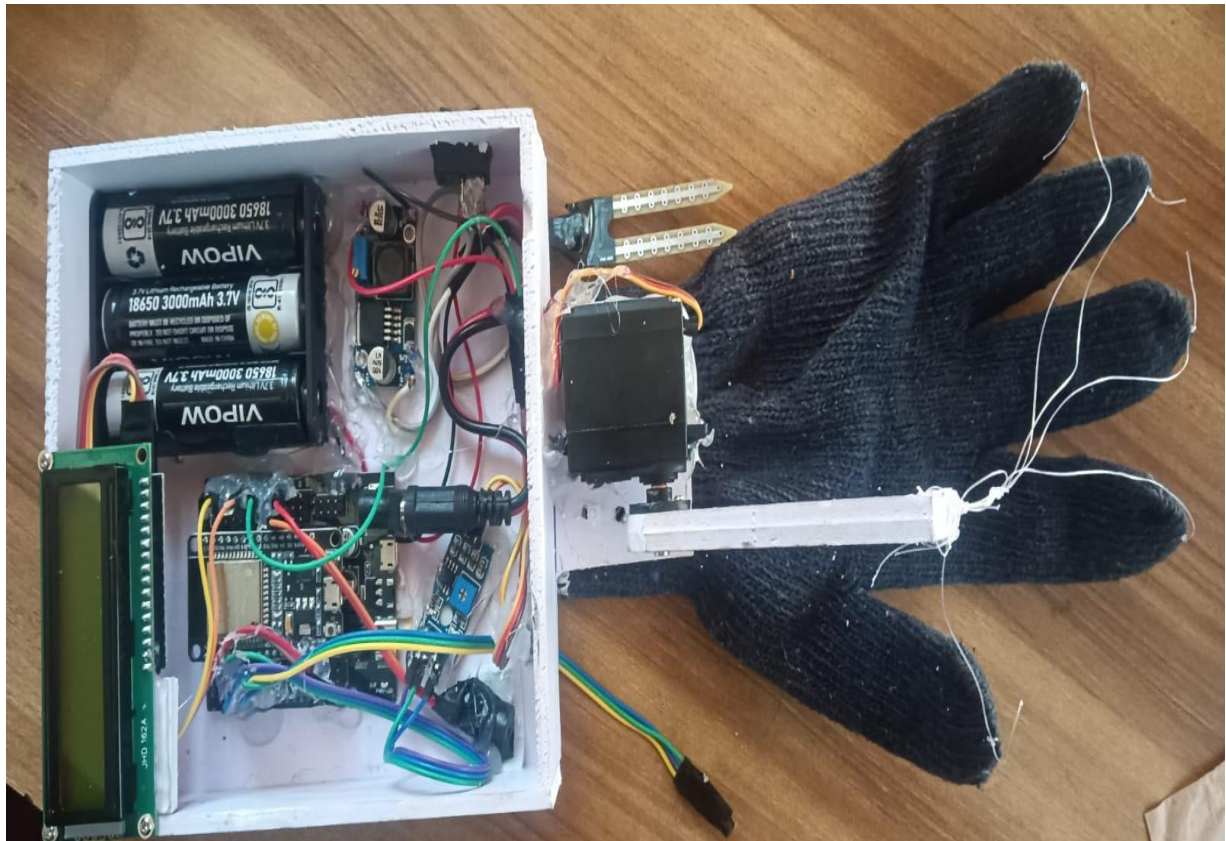


Figure 5: Prototype of the Smart Assistive Glove

CHAPTER 9 — FUTURE SCOPE

1. Full Hand Exoskeleton– Upgrade to a complete soft exoskeleton for independent finger control.
2. AI-Based Assistance– Use machine learning to predict patient gestures and assist movement automatically.
3. Enhanced Medical Monitoring– Add sensors for heart rate, SpO₂, blood pressure, and EMG for full health tracking.
4. Wireless Charging– Enable inductive charging for easier, cable-free use.
5. Mobile App Integration– Real-time monitoring, alerts, and rehabilitation logs through a smartphone app.
6. Voice Control Support– Allow patients to operate the glove using simple voice commands.
7. Cloud Data Storage– Save patient history online for doctor analysis and long-term tracking.
8. Soft Robotics– Replace servo motors with soft actuators for more comfort and flexibility.
9. Gesture-Based Control– Extend glove functions to control wheelchairs, home appliances, and assistive devices.
10. Rehabilitation Gaming– Add game-based physiotherapy to improve engagement and recovery
11. Solar Charging Option– Integrate thin solar sheets to extend battery life.
12. Medical Certification & Commercial Use– Upgrade the device to meet medical standards for commercialization.

CHAPTER 10: CONCLUSION

The Glove for Paralysis Patients is an innovative, low-cost, and practical assistive device designed to improve the quality of life for individuals with hand paralysis. By integrating sensors, servo-assisted movement, and IoT-based monitoring, the system provides real-time support, emergency alerts, and rehabilitation assistance in a compact wearable form. The glove enhances patient independence by enabling partial hand movement, communication assistance, and continuous health monitoring.

Its lightweight design, battery-powered operation, and customizable structure make it suitable for home use, hospitals, and physiotherapy centers. The project successfully demonstrates how embedded systems, wearable technology, and automation can work together to create an effective and user-friendly rehabilitation tool.

With future enhancements such as AI-based gesture prediction, mobile app integration, and full exoskeleton support, this system has strong potential for medical commercialization and broader adoption in healthcare. Overall, the project shows great promise in bridging the gap between patient needs and modern assistive technology.

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